DELHI PUBLIC SCHOOL JANKIPURAM, LUCKNOW

SESSION 2024-25

ANNUAL EXAMINATION

MATHEMATICS (041)

CLASS –IX

SAMPE PAPER-II

TIME:- 3 HOURS M.M.-80

General Instructions:

1. This Question paper contains - five sections A, B, C, D and E. Each section is compulsory. However, there is some internal choice in some questions.
2. Section A has 18 MCQ's and 02 Assertion Reason based questions of 1 mark each.
3. Section B has 5 Very Short Answer (VSA) questions of 2 marks each.
4. Section C has 6 Short Answer (SA) questions of 3 marks each.
5. Section D has 4 Long Answer (LA) questions of 5 marks each.
6. Section E has 3 source based/case based/passage based/integrated units of assessment (04 marks each) with sub parts.
7. Internal Choice is provided in 2 questions in Section-B, 3 questions in Section-C, 2 Questions in Section-D. You have to attempt only one alternative in all such questions.
8. If you wish to re-answer any question then, cancel the previous attempt first.
9. Attempt the questions in the order of best known to least known.

Q1. The value of $\frac{1}{\sqrt{10}}$ when $\sqrt{10} = 3.162$ is

a) .3162
b) 31.62
c) .03162
d) 316.2

Q2. The graph of the linear equation $2x + 3y = 6$ cuts the y-axis at the point:

a) (2, 0)
b) (0, 3)
c) (3, 0)
d) (0, 2)

Q3. Which point lies above x-axis?

(a) (-1, 2)
(b) (2, 0)
(c) (-1,-5)
(d) (0,-3)

Q4. If $x + y = 10$ then $x + y + z = 10 + z$. Then the Euclid's axiom that illustrates this statement is:

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a) First axiom
b) Second axiom
c) Third axiom
d) Fourth axiom

Q5. If the lateral surface area of a cylinder is 88 cm² and its height is 7 cm, what is its radius?
a) 2 cm
b) 3 cm
c) 4 cm
d) 5 cm

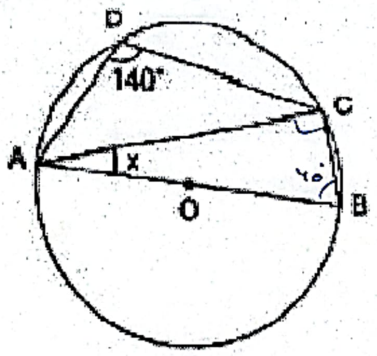
Q6. In a frequency distribution, the mid value of a class is 10 and the width of the class is 6. The lower limit of the class is:
a) 6
b) 7
c) 8
d) 12

Q7. The locus of a point which moves such that its distance from a fixed point is always constant is:
a) Line
b) Circle
c) Parabola
d) Ellipse

Q8. If two circles touch each other internally, the distance between their centers is:
a) Sum of their radii
b) Difference of their radii
c) Half the sum of their radii
d) Double the difference of their radii.

Q9. In the figure, O is the centre of the circle. If ∠ADC = 140°, then what is the value of x?

a) 30°
b) 50°
c) 60°
d) 40°



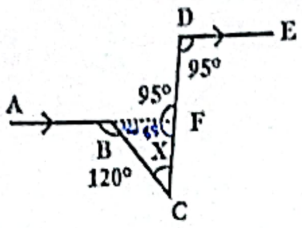
Q10. If an angle is one-third of its supplement, then the measure of the angle is
a) 45^0

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b) 30^0
c) 135^0
d) 150^0

Q11. Find the value of x, if AB is parallel to DE in the given figure:

a) 45°
b) 25°
c) 55°
d) 35°



Q12. The point where all medians of a triangle meet is called:

a) Orthocenter
b) Centroid
c) Incenter
d) Circumcenter

Q13. If the sides of a triangle are in the ratio 3:4:5, then the triangle is:

a) Equilateral
b) Isosceles
c) Scalene
d) Right-angled

Q14. If ABCD is a trapezium in which AB || CD and AD = BC, then:

a) $\angle A = \angle B$
b) $\angle A > \angle B$
c) $\angle A < \angle B$
d) None of the above

Q15. Two angles of a quadrilateral are 50° and 80° and other two angles are in the ratio 8 : 15. Find the measure of the remaining two angles.

a) 100°, 130°
b) 140°, 90°
c) 80°, 150°
d) 70°, 160°

Q16. The diagonals of rhombus are 12 cm and 16 cm. The length of the side of rhombus is:

a) 12 cm

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b) 16 cm
c) 8 cm
d) 10 cm

Q17. An isosceles right triangle has an estimated area $8cm^2$. The length of its hypotenuse is:

a) 16 cm
b) 48 cm
c) 32 cm
d) 24 cm

Q18. The sides of a triangle are 11 cm, 15 cm and 16 cm. The altitude to the largest side is

a) $30\sqrt{7}$cm
b) 30 cm
c) $15\frac{\sqrt{7}}{2}$ cm
d) $15\frac{\sqrt{7}}{4}$ cm

Q19 **Assertion:** If two interior angles on the same side of a transversal intersecting two parallel lines are in the ratio 5:4, then the greater of the two angles is 100^0
Reason: If a transversal intersects two parallel lines, then the sum of the interior angles on the same side of the transversal is 180^0

a) Both Assertion and reason are correct and reason is correct explanation for Assertion
b) Both Assertion and reason are correct but reason is not correct explanation for Assertion
c) Assertion is correct but reason is false
d) Both Assertions and reason are false

Q20. **Assertion:** The difference between the maximum and minimum values of a variable is called its range. **Reason:** The number of times a variate (observation) occurs in a given data is called range.

a) both Assertion and reason are correct and reason is correct explanation for Assertion
b) both Assertion and reason are correct but reason is not correct explanation for Assertion
c) Assertion is correct but reason is false
d) both Assertions and reason are false

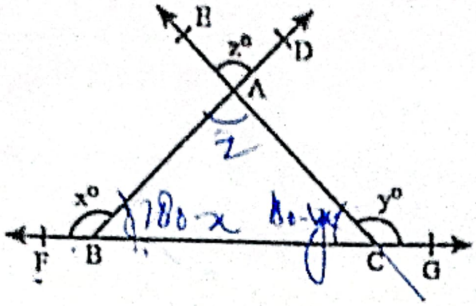
Section B

Q21 Find the value of

$$\frac{(\sqrt{48}+\sqrt{32})}{(\sqrt{27}+\sqrt{18})}$$

Q22. In figure, what is z in terms of x and y?

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$z = 180 - x + 180 - y$

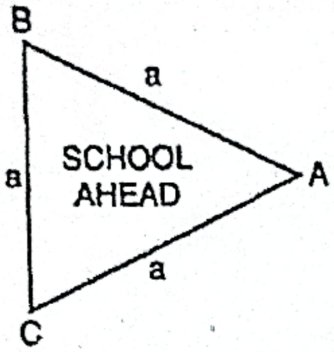
$180 = z + 180 - x + 180 - y$

$180 = 360 + z - x - y$

$z = -180 + x + y$

$\boxed{z = x + y - 180}$

Q23 A traffic signal board, indicating SCHOOLAHEAD is an equilateral triangle with side a Find the area of the signal board, using Heron's formula. If its perimeter is 180 cm, what will be the area of the signal board?



OR

Find the cost of laying grass in a triangular field of sides 50 m, 65 m and 65 m at the rate of Rs7 per m^2

Q24 ABC is an isosceles triangle in which AB = AC. Side BA is produced to D such that AD = AB. Show that ∠BCD is a right angle.

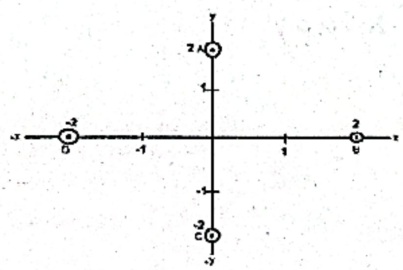
Q25. If one angle of a parallelogram is 30° less than twice the smallest angle, then find the measure of each angle.

OR

A diagonal of a rectangle is inclined to one side of the rectangle at 25°. Find acute angle between the diagonals. 65°

Section C

Q26.In fig. write the Co-ordinates of the points and if we join the points write the name of fig. formed. Also write Co-ordinate of intersection point of AC and BD.



OR

Write the mirror image of the given Point (2, 3)

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and (-4, -6) with respect to the x-axis.

Q27. Represent $1 + \sqrt{3}$ on the number line.

Q28. Write linear equation $3x + 2y = 18$ in the form of $ax + by + c = 0$. Also write the values of a, b and c. Are (4, 3) and (1, 2) solution of this equation?

OR

i) Write two solutions of the form $x = 0, y = a$ and $x = b, y = 0 : -4x + 3y = 12$

ii) Linear equation $y - 2 = 0$ is parallel to which axis.

Q29. An iron pillar consists of a cylindrical portion 2.8 m high and 20 cm in diameter and a cone 42 cm high is surmounting it. Find the weight of the pillar, given that 1 cm^3 of iron weighs 7.5 g.

OR

The radius of a cylinder is 7 cm. If its volume is 2002 cm^3, then find its height and total surface area.

Q30. If $x+y = 5$ and $xy = 4$. Find x-y by using identities.

Q31 Factorise

i) $16x^4 - y^4$

ii) $x^2+4y^2+9z^2-4xy-12yz+6xz$.

Section D

Q32. A rectangular piece of paper is 22 cm long and 10 cm wide. A cylinder is formed by rolling the paper along its length. Find the volume of the cylinder.

OR

A patient in a certain hospital is given soup daily in a cylindrical bowl of diameter 7cm. If the cylindrical bowl is served with soup up to a height of 4cm, how much soup ought to call the hospital to prepare daily to serve 250 patients? (Assume $\pi = 22/7$) 39.5 Litres

Q33.

i) Draw the graphs of $y = x$ and $y = -x$ in the same graph. Also find the co-ordinates of the point where the two lines intersect.

ii) Cost of pen is two half times the cost of a pencil. Express this situation as a linear equation in two variable $2.5x = y$

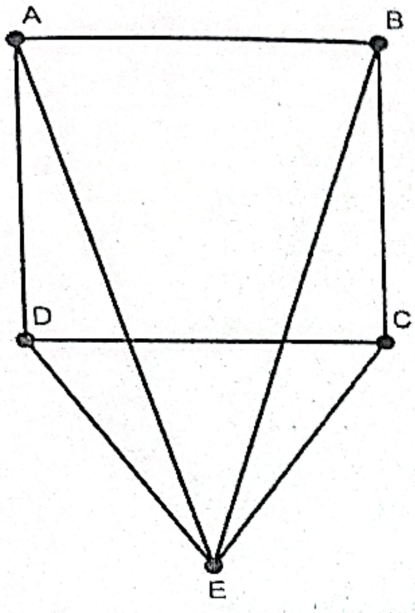
Q34. Find the values of a and b so that the polynomial $(x^4 + ax^3 - 7x^2 - 8x + b)$ is exactly divisible by $(x + 2)$ as well as $(x + 3)$.

OR

If $x^3 + ax^2 - bx + 10$ is divisible by $x^2 - 3x + 2$, find the values of a and b.

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Q35

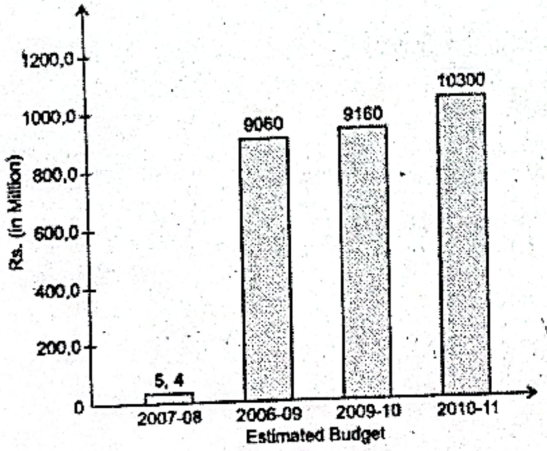


In the figure, ΔCDE is an equilateral triangle formed on a side CD of a square ABCD. Show that ΔADE≅ ΔBCE and also find ∠DAE

Section E

Q36. Read the following text carefully and answer the questions that follow:
Ladli Scheme was launched by the Delhi Government in the year 2008. This scheme helps to make women strong and will empower a girl child. This scheme was started in 2008.
The expenses for the scheme are plotted in the following bar chart.





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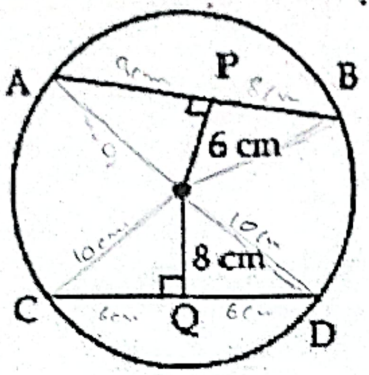
i. What are the total expenses from 2009 to 2011?
ii. What is the percentage of no of expenses in 2009-10 over the expenses in 2010-11?
iii. What is the percentage of mininum expenses over the maximum expenses in the period 2007-2011?

OR

What is the difference of expenses in 2010-11 and the expenses in 2006-09?

Q37. Read the following text carefully and answer the questions that follow:

Rohan draws a circle of radius 10 cm with the help of a compass and scale. He also draws two chords, AB and CD in such a way that the perpendicular distance from the center to AB and CD are 6 cm and 8 cm respectively.



Now, he has some doubts that are given below.

i. Show that the perpendicular drawn from the Centre of a circle to a chord bisects the chord.

ii. What is the length of CD?

iii. What is the length of AB?

OR

How many circles can be drawn from given three non collinear points?

Q38. Mrs. Rakhi lives in an undeveloped area where there is no facility of proper education. But one thing is available in that area i.e., network. Since she was very keen to take education, so she decided to complete her education through e-learning.

One day she was studying number system, where she learnt about rational numbers, irrational numbers and decimal numbers, etc.

On the basis of the above information, solve the following questions:

i) Write two rational between $\frac{1}{2}$ or $\frac{3}{4}$.

ii) Write rationaling factor of $\frac{1}{\sqrt{4}-\sqrt{3}}$

iii) If $\frac{\sqrt{3}-1}{\sqrt{3}+1}+\frac{\sqrt{3}+1}{\sqrt{3}-1}=a+\sqrt{3}\,b$. then find a and b.

OR

Prove that $(a^{-1}+b^{-1})^{-1}=\frac{ab}{a+b}$

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